Cross-cultural adaptation of the System Usability Scale into Afrikaans

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Abstract

We present AF-SUS, an Afrikaans cross-cultural adaptation of the System Usability Scale. Usability instruments are overwhelmingly administered in English, yet in multilingual settings such as South Africa, questionnaire wording may itself introduce variation unrelated to the system being evaluated. Rather than translating directly, we followed an iterative adaptation process and then examined the resulting instrument with Afrikaans-speaking participants who completed both versions across six everyday systems, followed by semi-structured interviews. Pretesting altered eight of the ten synthesised items, and interviewees understood the AF-SUS but consistently found the English version easier, with difficulty centring on individual terms in both languages; mean scores were similar across conditions (76.25 Afrikaans, 73.57 English), although the differing system sets and small sample make this descriptive rather than evidence of measurement equivalence.

CCS Concepts

• **Human-centered computing** → **Usability testing**; *Accessibility design and evaluation.*

Keywords

System Usability Scale, Human-Computer Interaction, multilingual usability, South Africa, Afrikaans, questionnaire evaluation

1 Introduction

Usability evaluations often rely on instruments that measure users' satisfaction with a system. One of the most widely used instruments for this purpose is the System Usability Scale (SUS), a ten-item questionnaire originally developed by John Brooke [9]. Its brevity, simplicity, and applicability across different types of systems have contributed to its widespread use in HCI research and practice.

The SUS was originally developed in English [9, 10], and its statements contain terminology and linguistic constructions that may not be equally accessible to users from different linguistic backgrounds. This is important because the SUS measures perceived usability: responses depend on participants being able to understand the statements sufficiently to express their perceptions of the system. Consequently, difficulties interpreting the questionnaire can potentially introduce variation that is unrelated to the usability of the system being evaluated. Previous research has identified terminology within the SUS that can be problematic for non-native English speakers, including the word "cumbersome" [14]. Research on cross-cultural usability evaluation has similarly shown that questionnaire adaptation may require consideration of linguistic and cultural differences rather than direct translation alone [12, 22].

This issue is particularly relevant in South Africa, where multiple languages are used in everyday communication and where English is not the home language of many users [23]. The SUS has previously been used in South African usability studies, including evaluations of online banking [19] and healthcare systems [13, 16, 17]. However, there is limited work examining how the SUS itself can be adapted for South African-language users. In particular, an Afrikaans version developed through a documented cross-cultural adaptation process would provide a basis for usability evaluations involving Afrikaans-speaking participants.

This study therefore investigates the cross-cultural adaptation of the SUS into Afrikaans. Afrikaans was selected because it is widely used in the Western Cape and is an official language of South Africa and the University of Cape Town [2], where the study was conducted. Rather than treating translation as a direct substitution of English words with Afrikaans equivalents, the study follows an iterative adaptation process involving two forward translations, expert synthesis, pretesting, participant-generated back-translations, and a final revision.

The resulting Afrikaans System Usability Scale (AF-SUS) is then examined using qualitative and exploratory quantitative evidence. Semi-structured interviews investigate how participants interpret and experience the two language versions, while SUS scores are analysed descriptively to identify broad differences in response patterns. Given the small sample, the quantitative component is intended as exploratory evidence rather than a psychometric validation of the instrument.

The study addresses the following research questions:

(1) **RQ1:** How can the System Usability Scale be cross-culturally adapted into Afrikaans for use with Afrikaans-speaking South Africans?
(2) **RQ2:** How do Afrikaans-speaking participants experience and interpret the English and Afrikaans versions of the System Usability Scale?
(3) **RQ3:** What exploratory differences in SUS scores are observed between the English and Afrikaans versions in this study?

The study makes three contributions. First, it provides a documented cross-cultural adaptation of the SUS into Afrikaans. Second, it identifies linguistic and interpretive issues encountered by Afrikaans-speaking participants when using the adapted instrument. Third, it provides preliminary descriptive evidence concerning SUS responses across the two language versions and identifies considerations for subsequent validation studies.

2 Background

2.1 The System Usability Scale

The System Usability Scale (SUS) is a ten-item Likert-style questionnaire developed by John Brooke [9] as a general-purpose measure of perceived system usability. It was designed to provide a short, technology-independent measure that could be administered following interaction with a system. The SUS uses a five-point Likert scale and alternates positively and negatively worded statements, producing a SUS score between 0 and 100 after the responses are transformed and summed [9].

The SUS has subsequently become one of the most widely used standardised usability questionnaires for usability evaluations [5, 10], in part because of its brevity and applicability across different types of systems [3, 18]. Its widespread use has also resulted in translations of the SUS into various languages [10, 15].

However, some aspects of the questionnaire may be interpreted differently based on the respondent's cultural, social, and linguistic background [12, 22], even if the respondent speaks English [14]. Thus, in a multilingual context such as South Africa [23], there is a need for a culturally and linguistically sensitive adaptation of the SUS into local languages.

2.2 Language and Cross-Cultural Measurement

Questionnaires are not necessarily transferable between languages through word-for-word translation. Adaptations must go beyond mere literal translation and instead take into account cultural factors to ensure the same meaning as the original is conveyed [22].

Beaton et al. [4] provide guidelines for the cross-cultural adaptation of self-report measures such as the SUS. They outline a six-stage process consisting of translation, synthesis, back-translation, expert committee review, pretesting, and document submission.

Ortiz-Gutiérrez and Cruz-Avelar [21] provide a more simplified version of these guidelines by noting that deciding to form a large committee to review the back-translation as described in stage 4 above may lead to increased costs and prolong the process, and that the translators used in previous steps are sufficient for the review. The study further states that just 6–8 participants are sufficient for the piloting stage.

Pan and Fond [22] provide a coding scheme for identifying, categorizing, and correcting issues in translation which can be useful in evaluating the effectiveness of translations during the review and synthesis stages of the adaptation process. These categories are summarized in Table 1.

2.3 Research Gap

Although the SUS has been translated and evaluated in multiple languages, comparatively less attention has been given to the cross-cultural adaptation and linguistic appropriateness of the SUS for South African-language contexts. In particular, there is a need for empirical examination of how Afrikaans-speaking users interpret an adapted version of the instrument rather than assessing translation quality solely through linguistic or psychometric comparison.

This study addresses this gap by developing an Afrikaans adaptation of the SUS through an iterative translation and pretesting process and examining how bilingual English and Afrikaans-speaking participants understand and experience the resulting instrument. Participant feedback is considered alongside exploratory SUS scores to examine both the linguistic suitability of the adaptation and descriptive differences between the two language versions. The study therefore treats adaptation as a problem involving both preservation of the instrument's intended meaning and accessibility to its target population.

3 Methodology

3.1 Research design

This study employs a within-subjects, mixed-methods approach to answer its research questions. The study's iterative cross-cultural adaptation process is described below. The qualitative component consists of semi-structured interviews intended to investigate participants' experiences of the two versions, including difficulties in understanding particular statements. The quantitative component provides supplementary evidence of the instrument's performance by examining and comparing the descriptive statistics of the results obtained from the AF-SUS and the original English SUS.

A within-subjects design was selected because it allows each participant to use both versions of the instrument. This reduces the extent to which differences between participants influence comparisons between the two versions. However, within-subjects designs may introduce order and learning effects because participants encounter the SUS more than once. To reduce this effect, different systems were evaluated during the two stages of the study. The ramifications of this are discussed in Section 3.9 Quantitative Analysis.

3.2 Participants and Recruitment

3.2.1 Inclusion Criteria. Participants were included in the study if they were:

(1) Above 18 years of age;
(2) First-language Afrikaans speakers; and
(3) Proficient in English.

No formal English proficiency threshold was imposed. However, participants were expected to have sufficient English proficiency to complete the English SUS. A basic level of English proficiency was assumed, as recruitment materials were distributed largely in English.

For the purposes of the study, a first-language Afrikaans speaker was operationalized as follows: Participants who listed Afrikaans as a Home Language (possibly among other languages) and rated their proficiency as either native or fluent were included in this study.

The terms "first language" and "Home Language" were used in relation to the participant's self-reported language background. Home Language was used as the primary indicator because participants may use the term "first language" differently, particularly in multilingual contexts. However, identifying Afrikaans as a Home Language alone was not considered sufficient to establish proficiency; therefore, participants' self-reported Afrikaans proficiency was also considered when determining eligibility.

3.2.2 Sampling. Participants were recruited from the University of Cape Town and from the broader public. Recruitment involved

announcements through the Department of Student Affairs, Human Resources for access to UCT staff, public notices, and snowball sampling.

This recruitment strategy was intended to provide variation in participants' backgrounds rather than to produce a statistically representative sample of Afrikaans speakers in South Africa.

Participants who also took part in the pretesting phase of the adaptation process were not excluded from the data collection. The pretesting phase was only for reviewing the wording of the SUS statements and not obtaining the SUS scores themselves. Potential learning effects were mitigated by the fact that the wording changed partially between the pretesting and final versions.

3.3 Ethical Considerations and Data Management

3.3.1 POPIA. Microsoft Forms was used as the platform for participants to complete the surveys, with responses automatically recorded in the associated Microsoft Excel workbooks. A Microsoft List was also used as a participant tracker across the different forms. These services were accessed through the University of Cape Town's institutional Microsoft 365 environment. UCT's Privacy and Data Protection Policy applies to research participants and researchers and requires personal information to be appropriately protected, minimised, and securely managed in accordance with the Protection of Personal Information Act (POPIA) [1]. Participant information was therefore managed within UCT's institutional environment and access to the collected data was restricted to the researchers.

3.3.2 Third-party Services. WhatsApp was used as a third-party platform for communication with participants. No sensitive personal or research information was transmitted through WhatsApp. A WhatsApp Community was used rather than a conventional group chat to hide participants' phone numbers and membership information from other participants. Consequently, WhatsApp was primarily used for study communication, while participant and survey data were collected and stored using UCT's institutional Microsoft environment.

3.3.3 Ethical Clearance. Ethical clearance was obtained from the UCT Faculty of Science Research Ethics Committee (FSREC), with approval number SCI/02801/2026.

3.4 Language Choice

Afrikaans was selected as the target language because of its status as an official language of South Africa and its prevalence in the Western Cape, where the study was conducted. Additionally, Afrikaans is an official language of the University of Cape Town [2], where this study was conducted.

Afrikaans is used by a diverse population and encompasses different linguistic varieties and registers [8]. The study therefore treats Afrikaans-speaking participants as a heterogeneous population rather than assuming that a single form of Afrikaans represents all speakers. This was addressed by an intentional decision to use more neutral, understandable wording, and through including different communities of Afrikaans speakers throughout the adaptation, pretesting, and data collection processes.

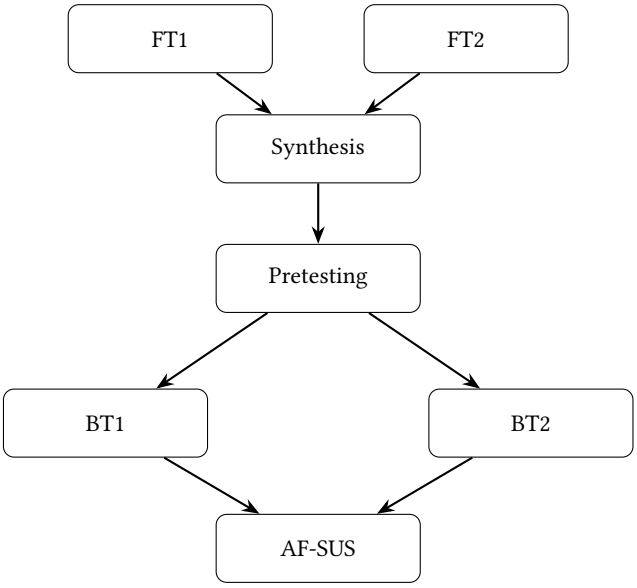


Figure 1: Cross-cultural adaptation process for the Afrikaans System Usability Scale (AF-SUS).

The researcher's own linguistic background was also relevant to the adaptation process. The lead researcher is a second-language Afrikaans speaker whose primary variety is associated with Cape Coloured Afrikaans, commonly referred to as Kaaps or Afrikaaps. This background may have influenced judgements concerning natural or appropriate Afrikaans wording during the adaptation process. This potential influence was therefore considered as part of the researcher's positionality.

3.5 Cross-cultural Adaptation

For the cross-cultural adaptation process, we set out to follow the guidelines of Ortiz-Gutiérrez and Cruz-Avelar [21], which were adapted from Beaton et al. [4] for low-resource contexts. An outline of the process is depicted in Figure 1.

3.5.1 Forward Translations. Firstly, two forward translations (English to Afrikaans) were produced. One was a researcher-produced translation made by the primary researcher (FT1) and one was a machine translation using Google Translate (FT2). A machine translation was used as they often generate more literal translations, while struggling with contextual accuracy [20]. The researcher-produced translation was done in a similar fashion, focusing on conveying the more literal meaning of each SUS statement from the original. This was done to provide two alternative formulations to be considered for synthesis.

3.5.2 Synthesis. These two translations were given to a linguistics expert for analysis and synthesis. The expert is an Afrikaans language professor specializing in language acquisition and translation/interpreting. They were asked to read the two forward translations, identify and categorise any errors in them, and finally to produce a synthesised version.

The errors were categorised according to the coding scheme outlined in Pan and Fond [22]. The scheme is outlined in Table 1.

Table 1: Coding Scheme for Identifying Issues in Translated Questionnaires. Adapted from Pan et al. [22]

Category	Description
Linguistic Rules	Identifies translation flaws related to improper lexical choice (obsolete terms), morphological errors, or unnatural syntax that inappropriately mirrors the source language's structure.
Cultural Norms	Addresses conceptual discrepancies where source language expressions (e.g., time, kinship, or address conventions) do not align with the target language's cultural standards.
Social Practices	Classifies untranslatable concepts that either do not exist in the target culture or lack a direct equivalent in the local social context.
Production Errors	Technical mistakes occurring during the survey creation phase, including typographical errors, formatting issues, or omitted text.
Respondent Errors	Refers to unintentional mistakes made by participants during the survey process, typically attributed to cognitive fatigue or inattention.

3.5.3 Pretesting. The synthesised version produced was then taken for pretesting. During this stage, participants were shown the instrument and asked to read the statements and provide feedback on what parts they struggled to understand, or which words they may have had to look up. The data collection for this phase took the form of an unstructured interview for which field notes were taken of responses. On top of this, two of the pretesting participants were asked to produce a back translation (Afrikaans to English) of the instrument.

Standard guidelines use professional back-translators and a second synthesis before pretesting. Our study instead combined these stages because of resource constraints, and used participants for the back-translations. The back-translations were therefore used as supplementary evidence of the target population's interpretation rather than as a formal translation validation.

Altogether, six people participated in the pretesting. The two back translations BT1 and BT2 were then passed along to the linguistics expert along with the summarised notes of the overall feedback from the participants.

3.5.4 Final Version. The expert then considered all the feedback and the different translations, and produced an updated final version of the System Usability Scale. This was the version used in the data collection stage of the study and had been named the AF-SUS. In particular, the expert considered the feedback and adjusted the translation to aim for maximum understanding across all the participants.

3.6 Qualitative Analysis

A semi-structured interview was conducted after the survey, with the participant's consent. The purpose of this interview was to investigate the interpretation of the AF-SUS and gain further insight into participants' understanding of the instrument, compared to the original English SUS.

After completing the survey, participants were asked whether they would like to participate in the interview stage. Interviews were conducted in English through WhatsApp, either as a phone call or asynchronously, with participants responding to the interview questions using text or voice notes. The asynchronous option allowed participants to complete the interview in their own time without scheduling a call. However, this format reduced opportunities for the interviewer to engage in spontaneous follow-up questions and clarification during the interview.

The interview data were analysed using Reflexive Thematic Analysis (RTA), following the approach established by Braun and Clarke [6, 7] and outlined by Byrne [11]. The analysis involved familiarisation with the responses, generation of initial codes, identification and development of candidate themes, and iterative refinement of the themes in relation to the research questions. The analysis was performed by one researcher. The researcher's linguistic background, discussed in Section 3.4, was considered during interpretation of the data.

3.7 Systems Under Evaluation

Six systems were selected for evaluation. These systems were selected to provide a context to use the SUS, rather than to obtain a definitive sample of usability ratings for each system. The systems selected are as follows:

(1) ATMs;
(2) Google Search;
(3) Uber;
(4) Takealot;
(5) POS machines; and
(6) EskomSePush.

ATMs and POS machines were treated as broad categories because participants may have interacted with multiple different makes/models of these technologies. The remaining systems were selected to represent different domains of everyday digital interaction.

Participants who were unfamiliar with a system were provided with a short task to establish sufficient context for evaluating it. Participants who had previous experience with a system were permitted to evaluate it based on their existing experience.

This has an effect on the evaluations as some participants are evaluating based on recent experience while others are evaluating from memory. This may lead to differing results as participants may have interacted with different versions of the same system, or be evaluating based on a different task.

3.8 Data collection

The participant pipeline for the study is depicted in Figure 2. Participants were first directed to a WhatsApp community for the study where an announcement was made instructing them to complete the demographic and consent form in their preferred language (English, Afrikaans, or isiXhosa).

They would then be individually contacted and asked to complete the English and Afrikaans surveys consisting of 3 systems each.

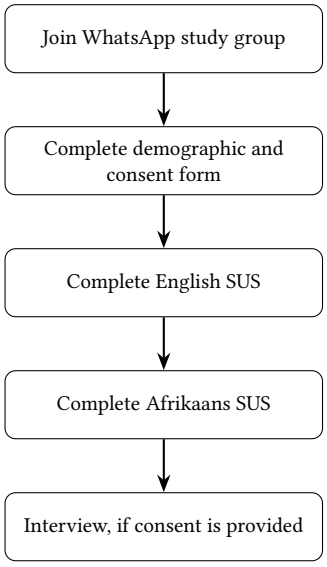


Figure 2: Participant procedure used for the main study.

Three systems were randomly picked from the 6 available for the English version, with the remaining 3 allocated to the Afrikaans version to have more diverse data for each instrument version.

Initially, the English and Afrikaans surveys were contained in two separate forms, with participants being contacted again to complete the second after completing the first. However, this was later changed to a combined English and Afrikaans form for ease of administration.

To ensure an equal distribution of system evaluations across languages, each participant was randomly assigned three systems to evaluate in English and three in Afrikaans. For the subsequent participant, the language assignments were reversed: the three systems previously assigned for English were assigned for Afrikaans, and vice versa. This alternating allocation ensured that, with an even number of participants, each system received an equal number of evaluations in each language.

Finally, participants were asked to engage in a short semi-structured interview where they share their experience. The participants were given the option to complete the interviews through a short call or asynchronously through voice or text messaging.

The management and administration of the study was conducted via WhatsApp. This was done so that participants could be sorted into different groups based on their native language(s), as this study was done in parallel to two other studies evaluating an isiXhosa version, and an orally-administered English version of the SUS respectively. Thus this provided a way to control the stratification of the participants and manage the two stages of the study, as well as the post-study interview. Additionally, the orally-administered SUS would be done over WhatsApp. An online procedure was chosen instead of in-person so that participants could complete each step of the process in their own time, reducing the time commitment of participants.

However, the use of WhatsApp, while it reduced logistical barriers, potentially favoured participants who had access to smartphones, mobile data, and who were familiar with WhatsApp.

3.9 Quantitative Analysis

The quantitative analysis examines the SUS scores obtained from the English SUS and the AF-SUS. The analysis compares descriptive statistics for the two versions to identify potential differences in the scores obtained. The quantitative results provide supplementary evidence alongside the qualitative findings.

3.9.1 SUS Scoring. SUS scores were calculated according to the method described by Brooke [9]. For each item, the response was converted to a value from 1 (Strongly Disagree) to 5 (Strongly Agree). For positively worded items (odd-numbered items), 1 was subtracted from the response value, while for negatively worded items (even-numbered items), the response value was subtracted from 5. The resulting item scores were summed and multiplied by 2.5, producing a total SUS score between 0 and 100.

3.9.2 Descriptive Analysis. Each participant evaluated three systems using the English SUS and three different systems using the AF-SUS. At the participant level, the three SUS scores obtained for each language were averaged to produce an average English SUS score and an average AF-SUS score for each participant. These averages were then used to compare the distributions of scores obtained from the two versions.

Although the same participants completed both versions, the systems evaluated under each language condition were different. Consequently, the participant-level scores do not represent paired measurements of the same systems. A direct paired comparison could therefore not isolate differences attributable to the language of the instrument. Differences between the two sets of scores may instead reflect differences between the systems evaluated, the order in which the versions were administered, the language of the instrument, or a combination of these factors.

In addition to participant-level analysis, system-level averages were calculated separately for each language. For each system, the SUS scores from all participants who evaluated that system were averaged. This provided a comparison of SUS scores at the level of individual systems and allowed potential differences between systems to be considered alongside the overall language-level comparison.

Descriptive statistics, including the mean, median, standard deviation, and range, were calculated for the participant-level scores and system-level scores. The overall means of the English SUS and AF-SUS were also compared.

4 Results

4.1 Cross-cultural adaptation

From the cross-cultural adaptation process firstly two forward translations were produced (Table 6). The linguistics expert then identified and classified the errors in each translation. They were classified according to the Coding Scheme by Pan and Fond [22] shown in Table 1.

FT1 resulted in 6 errors while FT2 resulted in 5. These are shown in Table 7. In the synthesised version, 5 statements retained the

same wording from either FT1 or FT2, while 5 were new translations determined to better convey the meaning of the original statement.

4.2 Pretesting

Six participants were recruited to pretest the synthesised Afrikaans translation. The sample comprised three women and three men. Two of these participants, one woman and one man, were additionally selected to produce back-translations of the synthesised translation.

Field notes from the pretesting interviews identified several instances of lexical and stylistic difficulty. Two participants were unfamiliar with the word *stelsel*, three were unfamiliar with *omslagtig*, and one was unfamiliar with *vermoë*. Two participants also commented that the wording was overly formal. These findings indicated that, although the synthesised translation was generally understandable, several lexical choices and aspects of its register could reduce accessibility to the intended respondents.

The back-translations provided further evidence of potential comprehension problems. The translator for BT1 initially did not understand *stelsel* or *omslagtig*, requiring their meanings to be explained. The participant also interpreted the phrase *aan die gang kon kom* as referring to the literal start-up of a system and suggested *met hierdie stelsel kon begin* as an alternative. The other back-translator for BT2 was familiar with *teenstrydigheid*, but nevertheless reported difficulty understanding the meaning conveyed by the corresponding statement. The synthesised version is shown alongside the back-translations in Table 8.

The field notes and back-translations were subsequently provided to the language expert, who used these findings to revise the synthesised translation. The resulting updated translation is presented alongside the initial synthesis in Table 2. Eight of the ten items were modified. Items 2 and 3 remained unchanged, while items 1 and 5 underwent only minor lexical changes. The remaining items underwent more substantial revisions, incorporating changes informed by the comprehension difficulties and stylistic concerns identified during pretesting.

Item 8 contained both *ingewikkeld* and *moeilik* as options for the translation of *cumbersome*. We decided to use *moeilik* in the final AF-SUS as it was thought to be more widely understood.

4.3 Participant details

The final sample size for the study was 7. The gender and self-reported English proficiency of the participants is shown in Figure 3. Of the expected 42 SUS evaluations, 39 were completed, as one participant did not provide ratings for the three Afrikaans systems. Three of the seven participants also took part in the pretesting.

4.4 Qualitative Analysis

Following the semi-structured post-survey interviews, reflexive thematic analysis (RTA) was performed. In total, 3 participants were interviewed in this stage, and have been labelled P1–P3. One participant responded via text on WhatsApp (P1), while the other two conducted the interview over a phone call.

During the RTA process, initial codes were developed, after which the codes were refined, candidate themes were developed, and final themes were then chosen. While the process was not

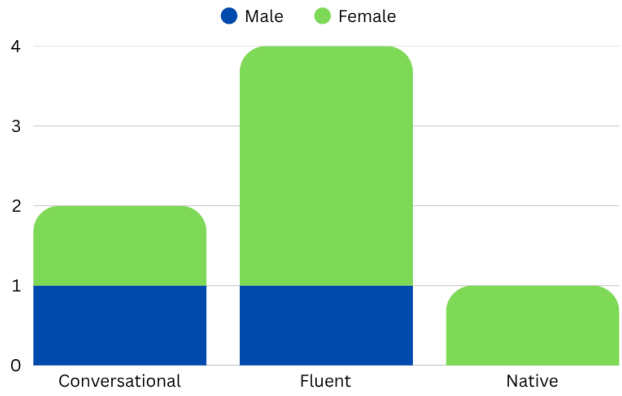


Figure 3: Number of participants in Data Collection by Self-reported English Proficiency

linear, the overall progression of the codes and themes are shown in Table 9 of the Supplementary Materials. The following four final themes were developed:

4.4.1 Comprehensibility of Afrikaans Translation. All participants noted that the Afrikaans translation was more difficult to understand than the original English. One participant noted that there were words that they "couldn't recognise in Afrikaans, but [...] knew the meanings of it, even though I didn't understand the word, but I knew what it was for." (P2) One particular word, pointed out by both P2 and P3 as being difficult to understand, was the Afrikaans word *inkonsekwenthede* in item 6 of the AF-SUS, roughly corresponding to the English word *inconsistencies*. One participant claimed that all statements were understandable, but still noted that the Afrikaans version "took a bit longer to understand compared to the English questions." (P1)

One participant also implied that the Afrikaans instrument was easier to understand because they completed the English one first:

> You could understand if you did the English one, you could understand basically what the Afrikaans one would be. (P2)

4.4.2 Relationship Between Language Versions. Participants noted that both versions of the survey seemed to ask the same thing. Participants also noted that differences in their answers were due to the different systems being evaluated, and not due to the language of the instrument itself. P3 states, after being asked whether they answered any items differently between the two versions:

> [...] most of the time, depending on the app being used, probably. But I'm pretty sure it would have been the same answer whether it was English or Afrikaans. (P3)

4.4.3 Language preferences and acceptability. Two participants stated that they would prefer to complete the English survey in a usability study. The other participant said they would prefer to do the Afrikaans version, but only to "challenge [themselves] more and engage more with the language on an educational level."(P1)

Table 2: Synthesised translation and final AF-SUS.

Item	Synthesised translation	Updated translation
1	Ek dink ek wil graag hierdie stelsel gereeld gebruik.	Ek dink ek sal graag hierdie stelsel gereeld wil gebruik.
2	Ek het die stelsel onnodig kompleks gevind.	Ek het die stelsel onnodig kompleks gevind.
3	Ek het gedink die stelsel is maklik om te gebruik.	Ek het gedink die stelsel is maklik om te gebruik.
4	Ek dink dat ek die ondersteuning van 'n tegniese kenner sou nodig hê om hierdie stelsel te kan gebruik.	Ek dink ek sou 'n tegniese kenner se hulp nodig hê om die stelsel te kan gebruik.
5	Ek het gevind dat die verskillende funksies in hierdie stelsel goed geïntegreer is.	Ek het gevind dat die verskillende funksies in die stelsel goed geïntegreer is.
6	Ek het gedink daar was te veel teenstrydigheid in hierdie stelsel.	Ek het gedink daar was te veel inkonsekwenthede in die stelsel.
7	Ek sou my voorstel dat die meeste mense baie vinnig sou leer hoe om hierdie stelsel te gebruik.	Ek sou dink dat die meeste mense gou sou leer hoe om die stelsel te gebruik.
8	Ek het die stelsel baie omslagtig gevind om te gebruik.	Ek het die stelsel baie moeilik gevind om te gebruik.
9	Ek het baie vertroue in my vermoë gehad toe ek die stelsel gebruik het.	Ek kon die stelsel met groot vertroue gebruik.
10	Ek moes baie dinge leer voordat ek met hierdie stelsel aan die gang kon kom.	Ek moes baie dinge leer voordat ek met die stelsel kon begin.

4.4.4 Comprehensibility of the Instrument. Participants described the statements as clear overall, although this may reflect comprehension achieved after looking up unfamiliar words. P3 stated that they did not understand the word "cumbersome" on the original English SUS.

4.5 SUS Scores

A total of 39 SUS scores were obtained. Six participants completed all six evaluations; one completed only the three English evaluations.

The overall statistics per language are shown in Table 3.

Table 3: Overall SUS Score Statistics by Language

Language	Count	Mean	Median	Std Dev	Min	Max	Range
Afrikaans	18	76.25	78.75	14.17	50.0	95.0	45.0
English	21	73.57	75.00	16.37	42.5	95.0	52.5
Total	**39**	**74.81**	**77.50**	**15.25**	**42.5**	**95.0**	**52.5**

The mean, standard deviation, median, and range for each system per language are summarized in Table 4. As stated in Section 3.8, there was an attempt to ensure that the number of evaluations of each system were consistent across languages, however this was not ultimately obtained due to participants choosing to rate different systems or not completing evaluations.

Table 5 further shows participant-level means between the English and Afrikaans versions.

5 Discussion

5.1 Principal Findings

This study investigated the cross-cultural adaptation of the System Usability Scale into Afrikaans and explored how Afrikaans-speaking participants experienced the resulting instrument. Three findings are particularly important. First, the adaptation process required changes beyond literal translation, with pretesting and post-survey interviews identifying terminology and phrasing that were not immediately accessible to some participants. Second, participants generally understood the intended meaning of the AF-SUS, but all three interviewees described the English version as easier to understand. Third, the English and Afrikaans conditions produced broadly similar descriptive SUS scores, although the study design prevents these observations from being interpreted as evidence of measurement equivalence or a causal language effect.

5.2 Cross-cultural Adaptation

The adaptation process demonstrates the value of iterative evaluation when translating usability questionnaires. Both forward translations contained linguistic issues, suggesting that literal translation strategies, whether human or machine-generated, are insufficient on their own. The pretesting phase then further suggested that an instrument that may be semantically equivalent to the original may still need to be further adapted to suit the particular context of the target users. The final version therefore incorporated changes based not only on translation and expert judgement but also on evidence from members of the target population.

This is consistent with the broader literature on cross-cultural adaptation, which distinguishes adaptation from literal translation by emphasising semantic, idiomatic, experiential, and conceptual equivalence [4, 22]. In this study, all identified issues were linguistic rather than cultural. This does not imply that cultural considerations were irrelevant, but rather that they were primarily expressed through the accessibility and naturalness of Afrikaans terminology.

5.3 Participant Experience of the Two Versions

The qualitative findings indicate a distinction between understanding the intended meaning of a statement and finding the statement immediately accessible. Participants generally reported that the AF-SUS statements were clear, yet all three interviewees considered

Table 4: SUS Score Statistics by System and Language

System	Afrikaans				English			
	Count	Mean	Std Dev	Median	Count	Mean	Std Dev	Median
ATM	3	79.17	1.44	80.00	4	63.12	23.84	62.50
EskomSePush	3	69.17	17.74	72.50	3	66.67	12.58	65.00
Google Search	3	89.17	5.20	87.50	4	80.62	20.65	88.75
POS Machine	3	71.67	21.26	72.50	4	80.00	10.00	75.00
Takealot	3	68.33	20.21	65.00	3	76.67	5.20	75.00
Uber	3	80.00	4.33	77.50	3	73.33	19.09	77.50

Table 5: Mean SUS Scores per Participant by Language

Participant	English Mean	Afrikaans Mean	Difference
P016	81.67	88.33	−6.67
P025	87.50	N/A*	N/A
P036	87.50	85.00	+2.50
P073	45.00	50.00	−5.00
P074	75.00	74.17	+0.83
P089	67.50	78.33	−10.83
P090	70.83	81.67	−10.83

* P025 completed only the English surveys and did not complete the Afrikaans survey.

the English version easier to understand. Difficulties centred on individual Afrikaans terms rather than on an inability to understand the underlying question.

This finding suggests that an adapted instrument can preserve broad semantic meaning while still requiring further refinement to achieve linguistic accessibility. The distinction is particularly relevant in multilingual contexts, where bilingual participants may have developed greater familiarity with technical or digital terminology in English.

The finding that participants perceived the two versions as asking essentially the same questions provides preliminary qualitative support for semantic equivalence between the instruments. At the same time, participants' preference for English demonstrates that semantic equivalence alone may not guarantee equivalent user experience of the questionnaire. This is relevant to usability measurement because difficulty processing questionnaire wording can affect the effort required to produce a response.

An additional finding was that one participant experienced difficulty with the English term "cumbersome". This is consistent with previous research identifying the same term as problematic for non-native English speakers [14]. Thus, the linguistic accessibility problem is not unique to the Afrikaans adaptation. Rather, the study illustrates that usability questionnaires may contain terminology that is familiar to researchers but less accessible to respondents in either language.

In the final AF-SUS, two of the three interviewed participants noted "inkonsekwenthede" as causing confusion. This word was introduced in the final version, to replace "teenstrydigheid" following pretesting feedback. This indicates that the revised version still produces issues, and more work may be needed to ensure greater comprehension.

5.4 Exploratory SUS Results

The overall mean SUS score was 73.57 for the English condition and 76.25 for the Afrikaans condition. The observed difference of 2.68 points is small relative to the variability of the scores. However, these descriptive similarities should not be interpreted as evidence that the two instruments are psychometrically equivalent.

The principal limitation is that participants evaluated different systems in the two language conditions. Consequently, differences in SUS scores may reflect differences in the usability of the systems being evaluated, rather than differences caused by the language of the questionnaire. The system-level results illustrate this variability, with some systems receiving higher scores in Afrikaans and others receiving higher scores in English.

Among the six participants who completed both language conditions, four had a higher mean score in the Afrikaans condition and two had a higher mean score in the English condition. However, these comparisons remain exploratory because the underlying system sets differed between conditions. The quantitative findings therefore provide useful descriptive evidence for subsequent research, but do not constitute a test of language equivalence, reliability, or validity.

5.5 Implications

Taken together, the findings suggest that future versions of the AF-SUS should prioritise terminology that is widely understandable across Afrikaans-speaking communities rather than favouring highly formal or less commonly encountered vocabulary. The adaptation process also suggests that participant feedback can identify problems that are not apparent during expert translation alone.

The findings further raise the possibility that multilingual usability evaluation may benefit from offering respondents a choice of language. Participants in this study did not uniformly prefer Afrikaans, despite considering Afrikaans a home language. A multilingual approach may therefore be more appropriate than assuming that a participant's home language is necessarily their preferred language for completing a usability instrument.

5.6 Limitations

The principal limitation is the small sample size. Seven participants contributed SUS data and only three participated in the post-survey interviews. The quantitative sample is therefore insufficient for psychometric validation or reliable estimation of population-level differences.

The allocation of different systems between language conditions further limits quantitative interpretation. Although participants completed both language versions, they did not evaluate the same systems in both conditions. The study therefore does not provide paired observations of the same system under different questionnaire languages. This was intended to reduce learning effects and ensure that participants re-evaluated statements in each language, however it introduced a confound in the analysis of the SUS scores.

The sample was also recruited through convenience and snowball-based methods and should not be regarded as representative of Afrikaans-speaking South Africans. Participants' English and Afrikaans proficiency was self-reported, and participants may differ in their familiarity with both the evaluated systems and the terminology used in them.

Finally, the interviews were conducted in English, and one interview was conducted asynchronously through text responses. These choices reduced logistical barriers but may have limited opportunities for spontaneous clarification and follow-up. The qualitative findings should consequently be understood as exploratory interpretations rather than a comprehensive account of Afrikaans-speaking users' experiences.

5.7 Future Work

A larger study should evaluate the AF-SUS with a substantially broader and more diverse sample of Afrikaans speakers. Future validation should examine internal consistency, reliability, construct validity, and, where appropriate, measurement equivalence with the original English SUS.

A stronger experimental design would also require participants to evaluate the same systems under both language conditions, ideally with counterbalanced questionnaire order. This would allow language-related differences to be separated more effectively from differences in system usability and order effects.

Further qualitative work should investigate participants from different Afrikaans-speaking communities and linguistic backgrounds, with particular attention to differences in terminology familiarity and preferred language for technology use. Similar adaptation processes could subsequently be applied to other South African languages, allowing comparison of methodological challenges across languages.

6 Conclusion

This study investigated the cross-cultural adaptation of the System Usability Scale into Afrikaans and explored how Afrikaans-speaking participants experienced the resulting instrument. The adaptation process demonstrated that translating SUS required more than replacing English terms with Afrikaans equivalents. Two forward translations, expert synthesis, target-population pretesting, participant-generated back-translations, and subsequent revision identified and addressed several issues involving terminology, formality, and sentence construction.

The qualitative findings indicate that participants generally understood the intended meaning of the AF-SUS, but consistently experienced the English version as easier to understand. This suggests that the adaptation achieved broad semantic comprehensibility while leaving some differences in linguistic accessibility. Importantly, participants also identified terminology in the original English SUS that was difficult to understand, indicating that accessibility issues are not unique to the Afrikaans adaptation.

The exploratory quantitative results showed similar overall SUS score distributions, with mean scores of 73.57 for English and 76.25 for Afrikaans. However, the small sample and the use of different systems in the two language conditions prevent these observations from being interpreted as evidence of measurement equivalence or validity.

The principal contribution of the study is therefore a documented Afrikaans adaptation process and preliminary evidence concerning how Afrikaans-speaking users interpret the resulting instrument. The AF-SUS provides a basis for future usability research, but its reliability, validity, and measurement equivalence should be established through a larger study using a design that controls more effectively for system and participant effects.

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AI Declaration

Generative AI, in particular ChatGPT, was used for proofreading and providing feedback on the structure of the paper. Feedback was used to identify inconsistencies, spelling errors, and grammar errors; all resulting edits were written by the author.

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A Supplementary Materials

Table 6: Forward translations of the SUS items

Item no.	Forward Translation 1	Forward Translation 2
1	Ek dink ek sal dit waardeer om die stelsel dikwels te gebruik.	Ek dink ek wil graag hierdie stelsel gereeld gebruik.
2	Ek het die stelsel onnodig kompleks gevind.	Ek het die stelsel onnodig kompleks gevind.
3	Ek het gedink die stelsel was maklik om te gebruik.	Ek het gedink die stelsel is maklik om te gebruik.
4	Ek dink ek sou ’n tegniese persoon se hulp nodig hê om die stelsel te kan gebruik.	Ek dink ek sal die ondersteuning van ’n tegniese persoon nodig hê om hierdie stelsel te kan gebruik.
5	Ek het die verskillende funksies goed geïntegreer gevind.	Ek het gevind dat die verskillende funksies in hierdie stelsel goed geïntegreer is.
6	Ek het gedink daar was te veel teenstrydighede in die stelsel.	Ek het gedink daar is te veel teenstrydigheid in hierdie stelsel.
7	Ek sou my voorstel dat die meeste mense gou sou leer hoe om die stelsel te gebruik.	Ek sou dink dat die meeste mense baie vinnig sal leer om hierdie stelsel te gebruik.
8	Ek het die stelsel baie omslagtig gevind om te gebruik.	Ek het die stelsel baie omslagtig gevind om te gebruik.
9	Ek het baie selfversekerd gevoel toe ek die stelsel gebruik het.	Ek het baie selfversekerd gevoel om die stelsel te gebruik.
10	Ek moes baie dinge leer voordat ek die stelsel kon gebruik.	Ek moes baie dinge leer voordat ek met hierdie stelsel kon begin.

Table 7: Errors identified in the two forward translations.

Translation	Item	Category	Description
FT1	1	Linguistic rules	“sal dit waardeer” is unnatural.
FT1	4	Linguistic rules	“support” does not mean “hulp”; “would need” is not always “sou nodig hê”; “tegniese kenner” is preferable to “tegniese persoon”.
FT1	5	Linguistic rules	The translation is too literal and direct and does not convey the meaning of the original sentence.
FT1	7	Linguistic rules	“leer hoe om” is not equivalent to “learn to use”: the inclusion of “hoe” (“how”) introduces another dimension to the question.
FT1	9	Linguistic rules	“selfversekerd” is not the best translation for “confident”, as it refers to a person’s personality or their general confidence.
FT1	10	Linguistic rules	“get going with the system” conveys more than “die stelsel gebruik”. It also implies an initial phase of becoming familiar with the system, which is not captured by the translation.
FT2	4	Linguistic rules	“support” has a broader meaning than “ondersteuning”; “I would need” is hypothetical, whereas “sou nodig hê” is more direct; syntactically, the use of “dat” is more appropriate in Afrikaans.
FT2	6	Linguistic rules	“is” is not appropriate, as the questions reflect on the use of the systems.
FT2	7	Linguistic rules	“would” is translated as “sal dink” and “sal leer”, changing the hypothetical concept to a certainty. “I would imagine” is not equivalent to “Ek sou dink”.
FT2	9	Linguistic rules	“selfversekerd” is not the best equivalent for “confident”, as it refers to personality or a person’s general confidence. The infinitive construction is unnatural, and there is a slight semantic difference between the English question and the translation.
FT2	10	Linguistic rules	“kon begin” is not equivalent to “get going”. “begin” is too literal.

Table 8: Synthesised translation and back translations.

Item	Synthesised translation	Back translation 1	Back translation 2
1	Ek dink ek wil graag hierdie stelsel gereeld gebruik.	I think I would like to use this system regularly.	I think I want to use this system regularly.
2	Ek het die stelsel onnodig kompleks gevind.	I found the system unnecessarily complicated.	I think the system was unnecessarily complex.
3	Ek het gedink die stelsel is maklik om te gebruik.	I thought the system was easy to use.	I thought the system was easy to use.
4	Ek dink dat ek die ondersteuning van 'n tegniese kenner sou nodig hê om hierdie stelsel te kan gebruik.	I think I need technical support to use this system.	I think I would have needed the assistance of technical expert to use the system.
5	Ek het gevind dat die verskillende funksies in hierdie stelsel goed geïntegreer is.	I find the different functions to use the system are well integrated.	I found that the different functions were well integrated into the system.
6	Ek het gedink daar was te veel teenstrydigheid in hierdie stelsel.	I thought there were too many difficulties in this system.	I thought that there were too many obstacles in the system.
7	Ek sou my voorstel dat die meeste mense baie vinnig sou leer hoe om hierdie stelsel te gebruik.	I think that the majority of people would find this system easy to use.	I would think that a lot of people will quickly learn how to use this system.
8	Ek het die stelsel baie omslagtig gevind om te gebruik.	I found this system very cumbersome to use.	I found the system difficult to use.
9	Ek het baie vertroue in my vermoë gehad toe ek die stelsel gebruik het.	I had a lot of confidence in my ability to use the system.	I had a lot of faith in my abilities while using the system.
10	Ek moes baie dinge leer voordat ek met hierdie stelsel aan die gang kon kom.	I had to learn many things before I could start the system.	I had to learn a lot before I could use the system effortlessly.

Table 9: Refinement of qualitative codes and themes.

Initial code	Refined code	Candidate theme	Final theme
Afrikaans harder to understand than English	Greater difficulty understanding Afrikaans	Afrikaans Comprehensibility	Comprehensibility of Afrikaans translation
Afrikaans preference for real study	Preference for Afrikaans in study context	Language preferences and acceptability	Language preferences and acceptability
Afrikaans translation of English words harder to understand	Difficult Afrikaans terminology	Relationship between language versions	Relationship between language versions
Afrikaans wording too complicated	Difficult Afrikaans terminology	Afrikaans Comprehensibility	Comprehensibility of Afrikaans translation
Both languages asked same thing	Perceived similarity between instruments	Relationship between language versions	Relationship between language versions
Could understand Afrikaans survey because of English	English version helped Afrikaans comprehension	Relationship between language versions	Relationship between language versions
Couldn't recognise some Afrikaans words	Difficult Afrikaans terminology	Afrikaans Comprehensibility	Comprehensibility of Afrikaans translation
Couldn't recognise word but knew meaning	Difficult Afrikaans terminology	Afrikaans Comprehensibility	Comprehensibility of Afrikaans translation
Difference in answers only due to different systems	Perceived similarity between instruments	Relationship between language versions	Relationship between language versions
English is participant's primary language	Participant linguistic background	Participant/questionnaire context	*Excluded*
No difference in answers between two versions	Perceived similarity between instruments	Relationship between language versions	Relationship between language versions
No problems with words or statements in either language	Perceived clarity of instruments	Instrument comprehensibility	Comprehensibility of Instrument
No unclear statements	Perceived clarity of instruments	Instrument comprehensibility	Comprehensibility of Instrument
Only a few Afrikaans words were difficult but otherwise it was easy	Minor difficulty with Afrikaans terminology	Afrikaans Comprehensibility	Comprehensibility of Afrikaans translation
Problem word: cumbersome	Difficulty with English terminology	Instrument comprehensibility	Comprehensibility of Instrument
Problem word: inkonsekwenthede	Difficulty with Afrikaans terminology	Afrikaans Comprehensibility	Comprehensibility of Afrikaans translation
Self-reported low Afrikaans proficiency	Participant linguistic background	Participant/questionnaire context	*Excluded*
Suggests using more understandable wording	Suggestion of simpler terminology	Language preferences and acceptability	Language preferences and acceptability
Understood questionnaire as a usability evaluation	Understanding of questionnaire in study context	Participant/questionnaire context	*Excluded*
Understood questionnaire in context of project	Understanding of questionnaire in study context	Participant/questionnaire context	*Excluded*
Would do Afrikaans survey for learning purposes	Preference for Afrikaans for specific purpose	Language preferences and acceptability	Language preferences and acceptability
Would like a mixed Afrikaans/English survey	Suggestion of combined language version	Language preferences and acceptability	Language preferences and acceptability
Would prefer English survey	Preference for English version	Language preferences and acceptability	Language preferences and acceptability